

UAE REAL ESTATE ANALYTICS

Dubai Analytics

Chart Reference Guide

Every chart in the Dubai section of the platform: what it shows, where the numbers come from, exactly how each figure is calculated, how to read it, what it does not tell you, and how it was checked.

33 charts documented · 6 dashboard sections

Prepared for company review. Local development build — not deployed.

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1. About this document

This guide documents every graph in the Dubai section of the UAE Real Estate Analytics platform. It is written for management, analysts and business users — you do not need to read code to use it.

It is generated directly from the application. The same descriptions appear behind the ⓘ icon next to each chart on screen, and every table of figures in this document is computed from the Dubai datasets when the document is built. If the dashboard changes, this guide is rebuilt from it — the two cannot drift apart.

How each chart is documented

- What the graph is, and why it is used.
- Which dataset it reads — RAW, CLEANED or DERIVED — and the exact file and columns.
- How the data is prepared, and the precise calculation.
- Axes, legend, and which dashboard filters affect it.
- How to read it, and how to explain it to a client.
- What it tells you — and, just as importantly, what it does not.
- Its limitations, and how the calculation was validated.

Three rules this dashboard follows

Median over mean. Property prices are right-skewed — a few very large deals pull the average upward. Medians are used for every headline comparison.

Rate per square metre over total price. Total price mixes size with value. Rate per m² is the like-for-like comparison, and it is the measure to quote when asked whether prices are rising.

Association, never causation. Where the dashboard compares two groups of properties, it says so. It never claims that a feature causes a price difference.

2. The two Dubai datasets

Two Dubai datasets ship with the platform. Every chart states which one it uses, in a badge beside its title on screen and in its entry in this guide.

	RAW registry	CLEANED dataset
File	transactions.parquet	latest_combined_data.parquet
Transactions	1,762,262	818,838
Columns	47	85
Date range	1966-01-18 → 2026-08-06	2010-01-01 → 2026-08-06
Areas	258	69
Transaction types	Sales, Mortgages, Gifts	Sales only
Property types	Unit, Villa, Land, Building	Unit only
Property usage	All	Residential only

The cleaned file is the residential-unit SALES slice of the raw registry, restricted to 2010 onwards and enriched with engineered time parts, unit attributes, amenity flags and building / developer scoring.

Of the 928,489 residential-unit sales in the raw registry, **818,838 (88.2%)** carry through to the cleaned dataset, which adds 38 engineered columns — time parts, unit attributes, amenity flags and building and developer scoring.

Which dataset each chart uses

Data source	Charts	Which ones
CLEANED	32	Where the market is concentrated; Building price tier vs realised rate; Building grade vs realised rate; Monthly market activity; Annual volume; Quarterly pattern; Year-over-year growth; Seasonality; Busiest areas; Most expensive areas; Value map; Volume against price; Locality zones; Nearest metro station; Layout mix; Typical size by layout; Rate per m ² by layout; Off-plan vs existing; Transaction procedure; Size against price; How prices are moving; Off-plan vs existing pricing; How amenities relate to price — headline comparison; How amenities relate to price — like-for-like; Where the price points are; Sale price distribution; Rate per m ² distribution; Unit size distribution; Sale price by registration type; How the price distribution has changed; Why the raw amenity number is so much bigger; Why off-plan looks more expensive
DERIVED	1	Published price forecast

A note on the amenity analysis

Only the parking flag exists in the raw registry. The swimming-pool, balcony, elevator and metro flags are engineered fields present only in the cleaned dataset, so they cannot be re-derived from the raw file. The parking result was additionally validated against the raw registry — see Appendix A.

3. What was checked, and what was found

Six areas of the Dubai dashboard were reviewed against the datasets. This is what the review found.

Chart	Verdict	What was found
Year-over-year growth	Correct, one issue of interpretation	Each year is compared with the immediately preceding year. All 17 years 2010–2026 are present with no gaps. 2026 is a partial year (data ends 2026-08-06), so its -52.9% volume bar is not comparable; the like-for-like figure is -23.8%. A like-for-like panel was added. Negative years were verified as genuine and kept.
Rate per m ² by layout	Chart was unreadable — rebuilt	All nine layouts were drawn at one x-position. Rebuilt as one panel per layout on a shared scale, with transaction counts and a full quartile table. The previous caption ("smaller units cost more per m ² ") was tested against the data, found to be false, and removed.
How prices are moving	Correct, readability improved	Monthly medians are correct. Every month carries at least 814 transactions (median 2,437), so the jaggedness is genuine, not thin-sample noise. A 3-month centred rolling median was added as an optional view: the standard deviation of month-on-month change falls from 7.6% to 4.4%. Actual observations remain selectable and tabulated.
Off-plan vs existing	Correct, made explicit	Classification validated on the raw registry: exactly two values, no missing rows. Off-plan traded at a premium in 17 of 17 years. A premium/discount chart was added so the gap is stated rather than eyeballed.
Amenities vs price	Headline was misleading — rebuilt	Recalculated and decomposed. Controlling for area, layout, year and registration type, every headline figure collapses (see Appendix A). The dashboard now shows headline and like-for-like side by side, plus a composition explorer.
Where the price points are	Correct, filter effect explained	Bands validated on all three datasets: counts sum exactly to the row count with zero unassigned rows and no duplicate transaction identifiers. The empty ">10M" band was caused by the default price filter, not by the data — a warning now says so.

4. Chart reference

Every chart in the Dubai dashboard, in the order it appears on screen.

4.1 Insights

4.1.1 Where the market is concentrated

How transaction value is distributed across Dubai's areas.

In one sentence

A few Dubai areas take most of the money. This shows how few.

WHAT IS THIS GRAPH?

A Pareto (ABC) chart. Bars are the total value transacted in each area, tallest first; the line is the running cumulative share of the whole market.

WHY IS THIS GRAPH USED?

To show how concentrated the market is. If a handful of areas account for most of the money, that is where attention, stock and risk sit.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take every sale that passes your sidebar filters.
2. Add up the money spent in each area.
3. Line the areas up, biggest spender first.
4. Draw the running total as a line, so you can see where it reaches 80%.

WORDS USED ON THIS CHART

Pareto chart — Bars sorted biggest first, with a line showing the running total, so you can see how few items make up most of the whole.

Cumulative share — A running total, as a percentage. Where it crosses 80%, everything to the left of that point accounts for four-fifths of the market.

Total value — Every sale price in the group added together — how much money changed hands.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	area_name_en, actual_worth

DATA PREPARATION

Rows are filtered by the sidebar selection. No other transformation.

CALCULATION

For each area, `sum(actual_worth)`. Areas are sorted descending; the line is the cumulative sum divided by the grand total, as a percentage. The top 25 areas are drawn; the cumulative line uses those 25.

AXES AND LEGEND

Element	What it shows
X-axis	Area, ordered by total transaction value (highest first).
Y-axis	Total value transacted in the area (AED).
Secondary (right) axis	Cumulative share of total market value (%), capped at 105 for readability.
Legend	Blue bars = value per area. Amber line = cumulative share. The dotted guide marks the 80% level.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Read the bars left to right — each one is a whole area's transaction value.
- Follow the amber line to where it crosses 80%: everything to the left of that point makes up four-fifths of the market.
- A line that rises steeply means a very concentrated market.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

Dubai's residential market is not spread evenly. This chart shows how much of the money is changing hands in how few places.

WHAT THIS GRAPH TELLS US

- Which areas carry the market's value, and how few of them there are.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about profitability, yield or future performance.
- High total value can come from many cheap deals or few expensive ones — check the accompanying table for both.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.

VALIDATION PERFORMED

The cumulative share is checked to reach 100% across all areas, and the per-area totals reconcile with the table underneath.

4.1.2 Building price tier vs realised rate

Median rate per m² for each price tier carried in the dataset.

In one sentence

Do the buildings we label “premium” actually sell for premium prices?

WHAT IS THIS GRAPH?

A ranking of the dataset's own building **Price Tier** label against the rate per m² those buildings actually achieved.

WHY IS THIS GRAPH USED?

The tier label was assigned by the project's building-scoring work. This chart is the sanity check that it lines up with what buyers really paid.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take every sale in your selection that has a price-tier label.
2. Group the sales by that label.
3. For each label, find the middle price per square metre.
4. Draw one bar per label so the labels can be compared.

WORDS USED ON THIS CHART

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	Price Tier, meter_sale_price

DATA PREPARATION

Rows labelled **Unknown** are excluded. Tiers with fewer than 200 transactions in the current selection are dropped so a thin group cannot top the ranking.

CALCULATION

`median(meter_sale_price)` per tier.

AXES AND LEGEND

Element	What it shows
X-axis	Median rate per m ² (AED).
Y-axis	Price tier.
Legend	Single series; hover shows the number of transactions behind each bar.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Tiers should step up in order. If two tiers overlap, the label is not separating the market as intended.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

A check that our building tiers mean something in the market, not just on paper.

WHAT THIS GRAPH TELLS US

- Whether the tier labels are consistent with realised prices.

WHAT THIS GRAPH DOES NOT TELL US

- Why a building sits in a tier — the scoring inputs are not in this dataset.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- Tier is a supplied label, not something recomputed here.

VALIDATION PERFORMED

Cross-checked against the Building Grade chart beside it; both should rank in the same direction.

4.1.3 Building grade vs realised rate

Median rate per m² by the A+ to D building grade in the dataset.

In one sentence

Same check as the tier chart, using the A+ to D building grade.

WHAT IS THIS GRAPH?

The dataset's building **Grade** against the rate per m² actually achieved.

WHY IS THIS GRAPH USED?

Same purpose as the price-tier chart: verifying that a supplied quality label is reflected in transaction prices.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take every sale that carries a building grade.
2. Group them by grade.
3. For each grade, find the middle price per square metre.
4. Rank the grades so you can see whether A+ really beats D.

WORDS USED ON THIS CHART

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

DATA SOURCE

Field	Value
Source	CLEANED

Field	Value
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	Grade, meter_sale_price

DATA PREPARATION

Unknown excluded; grades with fewer than 200 transactions dropped.

CALCULATION

median(meter_sale_price) per grade.

AXES AND LEGEND

Element	What it shows
X-axis	Median rate per m ² (AED).
Y-axis	Building grade.
Legend	Single series; hover shows transaction counts.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Expect A+ at the top and D at the bottom. Any inversion is worth investigating in the grading method.

HOW TO EXPLAIN IT TO A CLIENT**In plain language**

Do higher-graded buildings actually sell for more? This answers it.

WHAT THIS GRAPH TELLS US

- Whether building grade tracks realised price.

WHAT THIS GRAPH DOES NOT TELL US

- How the grade was derived.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.

VALIDATION PERFORMED

Compared with the price-tier ranking, which uses a different label.

4.2 Trends

4.2.1 Monthly market activity

Transaction count and total value, month by month.

In one sentence

How busy the market was, month by month, and how much money moved.

WHAT IS THIS GRAPH?

How many sales completed each month and how much money that represented.

WHY IS THIS GRAPH USED?

Volume is the clearest measure of market activity, and it moves before prices do.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take every sale in your selection and note which month it happened in.
2. Count the sales in each month, and add up their prices.
3. Draw the count as the shaded area and the money as bars.
4. Add a three-month average line so the general direction is visible through the monthly wobble.

WORDS USED ON THIS CHART

Total value — Every sale price in the group added together — how much money changed hands.

Rolling median — For each month, take that month together with the one before and the one after, and use the middle value of the three. It calms the zig-zag without moving the level or hiding anything — the real monthly figures are still there.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	instance_date, year_month, actual_worth

DATA PREPARATION

Transactions are grouped by calendar month using `year_month`.

CALCULATION

Per month: `count(rows)`, `sum(actual_worth)`, and a centred 3-month rolling mean of the count.

AXES AND LEGEND

Element	What it shows
X-axis	Month.
Y-axis	Number of transactions.
Secondary (right) axis	Total value transacted (AED).
Legend	Blue area = transactions. Dotted amber = 3-month average. Teal bars = total value.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- The blue area is the raw monthly count; the dotted line smooths it.
- Bars and line moving apart means the average deal size is changing.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

How busy the market has been, month by month.

WHAT THIS GRAPH TELLS US

- The direction and pace of market activity.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about price levels — see the Price section.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- The final month (August 2026) is **incomplete** — only 2,157 transactions to 6 August versus about 10,800 in a full month, so the last point always falls sharply.

VALIDATION PERFORMED

Monthly counts sum to the filtered row count.

4.2.2 Annual volume

Transactions completed each year.

In one sentence

How many homes changed hands each year.

WHAT IS THIS GRAPH?

One bar per calendar year, showing how many sales were registered.

WHY IS THIS GRAPH USED?

The simplest long-run view of whether the market is growing or contracting.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take every sale in your selection.
2. Count how many happened in each calendar year.
3. Draw one bar per year.

WORDS USED ON THIS CHART

Partial year — A year the data does not cover in full. 2026 stops on 6 August, so eight months of it are being compared with somebody else's twelve.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	year, actual_worth

DATA PREPARATION

Grouped by the year column already present in the dataset.

CALCULATION

count (rows) per year.

AXES AND LEGEND

Element	What it shows
X-axis	Year.
Y-axis	Number of transactions.
Legend	Bars are coloured red when the year fell against the previous one.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Taller bar = more deals closed.

HOW TO EXPLAIN IT TO A CLIENT**In plain language**

Year by year, how many homes changed hands.

WHAT THIS GRAPH TELLS US

- The long-run growth of the Dubai residential-unit market.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about prices.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- The 2026 bar covers January–August only and will look artificially low.

VALIDATION PERFORMED

Yearly counts sum to the filtered row count.

4.2.3 Quarterly pattern

Transaction counts by year and quarter.

In one sentence

A calendar grid: which three-month stretches are busy, and whether that has changed over the years.

WHAT IS THIS GRAPH?

A grid of years against quarters, shaded by how busy each quarter was.

WHY IS THIS GRAPH USED?

Separates the long-run trend (down the grid) from any repeating within-year rhythm (across the grid).

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Put every sale into the year and quarter it happened in.
2. Count the sales in each year-and-quarter square.
3. Shade each square — darker means busier.

WORDS USED ON THIS CHART

Partial year — A year the data does not cover in full. 2026 stops on 6 August, so eight months of it are being compared with somebody else's twelve.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	year, quarter, actual_worth

DATA PREPARATION

Pivot of the filtered rows; no aggregation beyond counting.

CALCULATION

`count ( rows )` for each year × quarter cell.

AXES AND LEGEND

Element	What it shows
X-axis	Quarter (Q1–Q4).
Y-axis	Year.
Legend	Darker cell = more transactions.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Scan down a column to see how one quarter has changed over the years.
- Scan across a row to see the shape of a single year.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

Which parts of the year are busiest, and whether that has changed.

WHAT THIS GRAPH TELLS US

- Whether Dubai has a recurring seasonal pattern, and how it has changed.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about prices or deal size.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- 2026 Q3 is partial and Q4 is absent.

VALIDATION PERFORMED

Cell counts sum to the filtered row count.

4.2.4 Year-over-year growth

Change in volume and in median rate per m², each year against the one before it.

In one sentence

Did the market get busier or quieter than last year, and did prices per square metre rise or fall?

WHAT IS THIS GRAPH?

For every year, the percentage change against the **immediately preceding year** — never against a fixed baseline.

WHY IS THIS GRAPH USED?

This is the chart that answers “compared with last year, did pricing go up or down?”

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Count the sales in each year, and find each year's middle price per square metre.
2. Take one year and the year immediately before it.
3. Work out the change as a percentage of that previous year.
4. Repeat down the whole series, so every year is compared with its own predecessor — never with a fixed starting year.

WORDS USED ON THIS CHART

Year-over-year — This year compared with the year immediately before it — never with a fixed starting year. So 2025 is measured against 2024, and 2024 against 2023.

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

Partial year — A year the data does not cover in full. 2026 stops on 6 August, so eight months of it are being compared with somebody else's twelve.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	year, actual_worth, meter_sale_price

DATA PREPARATION

Years are sorted ascending. Every year from 2010 to 2026 is present in the data, so no year is compared across a gap.

CALCULATION

Growth (%) = (this year – previous year) ÷ previous year × 100, applied to the transaction count and to median(meter_sale_price) separately. Implemented as pct_change() on a year-sorted series, so 2020 is compared with 2019, 2021 with 2020, and so on.

AXES AND LEGEND

Element	What it shows
X-axis	Year (the first year has no bar — there is nothing to compare it with).
Y-axis	Change in transaction volume vs the previous year (%).
Secondary (right) axis	Change in median rate per m ² vs the previous year (%).
Legend	Bars = volume growth, teal when positive and red when negative. Dotted blue line = rate growth.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Bars above the zero line mean more deals than last year; below means fewer.
- The dotted line is the pricing answer: above zero, the market got more expensive per m² than the year before.
- The two do **not** have to agree — a busy year can be a flat pricing year.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

“Against last year, did the market get busier or quieter, and did prices per square metre rise or fall?” Bars answer the first, the dotted line answers the second.

WHAT THIS GRAPH TELLS US

- Whether pricing rose or fell against the previous year, and by how much.
- Whether activity and pricing are moving together or apart.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about **why** a year moved.
- Nothing about the level of prices — only the change.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- **2026 volume is not comparable.** It covers January–August only, which is why it shows about –53%. On a like-for-like January–August basis against 2025 the fall is about –24%. The panel beside the chart shows that comparison. The negative value is kept because it is arithmetically correct for the data present.
- Median rate is far less affected by a shorter year than volume is: the full-year 2026 rate change is –0.1% and the like-for-like figure is +1.1%.

VALIDATION PERFORMED

Recomputed independently in pandas: every year is compared with the immediately preceding year, all 17 years are present with no gaps, and each negative year was checked against its own volume and median to confirm it is genuine rather than a formula or ordering error. 2011 (–24.9%), 2015 (–26.2%), 2018 (–30.4%) and 2020 (–14.0%) are real market contractions and have been kept.

4.2.5 Seasonality

Which calendar months are busiest, pooled across all years.

In one sentence

Which months of the year Dubai tends to buy in, with every year stacked together.

WHAT IS THIS GRAPH?

Total transactions in each calendar month, added up across every year in the selection.

WHY IS THIS GRAPH USED?

To see whether Dubai has a genuine within-year rhythm.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Ignore which year a sale happened in — keep only the month.
2. Add up all the Januaries, all the Februaries, and so on.
3. Draw one bar per calendar month, with the twelve-month average as a dotted line.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	month, actual_worth

DATA PREPARATION

Grouped by the month number; all years pooled.

CALCULATION

count (rows) per calendar month. The dotted line is the mean across the twelve months.

AXES AND LEGEND

Element	What it shows
X-axis	Calendar month.
Y-axis	Transactions across all years in the selection.
Legend	Amber dotted line marks the monthly average.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Bars well above the dotted line are consistently busy months.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

The months when Dubai buys.

WHAT THIS GRAPH TELLS US

- Whether activity is seasonal.

WHAT THIS GRAPH DOES NOT TELL US

- Whether the pattern still holds recently — pooling hides change over time. Use the quarterly grid for that.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- Pooling across 2010–2026 gives recent, higher-volume years more weight.
- September–December carry one fewer year than January–August because 2026 is incomplete.

VALIDATION PERFORMED

Monthly totals sum to the filtered row count.

4.3 Geography

4.3.1 Busiest areas

The 15 areas with the most transactions.

In one sentence

The fifteen areas where the most homes change hands.

WHAT IS THIS GRAPH?

A ranking of Dubai areas by number of sales.

WHY IS THIS GRAPH USED?

Shows where the market is most liquid — where you can buy and sell easily.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take every sale in your selection.
2. Count how many happened in each area.
3. Show the fifteen busiest.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	area_name_en

DATA PREPARATION

Filtered rows only.

CALCULATION

count (rows) per area; top 15 shown.

AXES AND LEGEND

Element	What it shows
X-axis	Number of transactions.
Y-axis	Area.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Longer bar = more deals. This is activity, not value.

HOW TO EXPLAIN IT TO A CLIENT**In plain language**

Where Dubai is trading most often.

WHAT THIS GRAPH TELLS US

- Where the most transactions happen.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about price — a busy area can be affordable.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.

- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.

VALIDATION PERFORMED

Counts reconcile with the area detail table in the same section.

4.3.2 Most expensive areas

Highest median rate per m², areas with 300+ transactions only.

In one sentence

Dubai's most expensive addresses, measured per square metre so that big homes do not automatically win.

WHAT IS THIS GRAPH?

A ranking of areas by the median price per square metre achieved there.

WHY IS THIS GRAPH USED?

Rate per m² removes unit size from the comparison, so a studio area and a penthouse area can be compared fairly.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Group every sale by area.
2. Throw out any area with fewer than 300 sales, so one unusual deal cannot put an area at the top.
3. For each remaining area, find the middle price per square metre.
4. Show the fifteen highest.

WORDS USED ON THIS CHART

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	area_name_en, meter_sale_price

DATA PREPARATION

Areas with fewer than **300** transactions in the current selection are excluded, so a single unusual sale cannot top the ranking.

CALCULATION

`median(meter_sale_price)` per area; top 15 of those that clear the 300-transaction threshold.

AXES AND LEGEND

Element	What it shows
X-axis	Median rate per m ² (AED).
Y-axis	Area.
Legend	Hover shows how many deals sit behind each bar.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Longer bar = higher price per square metre.
- Always read the transaction count in the tooltip alongside the bar.

HOW TO EXPLAIN IT TO A CLIENT**In plain language**

Dubai's premium addresses, measured per square metre so that size does not distort the comparison.

WHAT THIS GRAPH TELLS US

- Which areas command the highest price per square metre.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about total spend — an expensive area can be small.
- Nothing about value for money or investment return.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- The 300-transaction floor deliberately hides small, very expensive areas.

VALIDATION PERFORMED

Thresholded and unthresholded rankings were compared; the threshold removes areas whose median rested on a handful of deals.

4.3.3 Value map

Box size = total transaction value; colour = median rate per m².

In one sentence

One picture of where Dubai's money goes, and how pricey it is there.

WHAT IS THIS GRAPH?

A treemap of the top 25 areas by money transacted, shaded by price level.

WHY IS THIS GRAPH USED?

Combines two questions in one picture: where the money is, and whether it is expensive money or high-volume money.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Add up the money spent in each area.
2. Give each area a rectangle sized by that total.
3. Colour the rectangle by the area's middle price per square metre.

WORDS USED ON THIS CHART

Treemap — Nested rectangles where the size of each box is proportional to the number it represents.

Total value — Every sale price in the group added together — how much money changed hands.

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	area_name_en, actual_worth, meter_sale_price

DATA PREPARATION

Top 25 areas by total value.

CALCULATION

Box area $\propto$ `sum(actual_worth)`; colour = `median(meter_sale_price)`.

AXES AND LEGEND

Element	What it shows
Legend	Darker = higher rate per m ² .

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Large and dark = lots of money at high prices per m².
- Large and pale = high volume at accessible prices.

HOW TO EXPLAIN IT TO A CLIENT**In plain language**

One picture of where the money goes and how expensive it is there.

WHAT THIS GRAPH TELLS US

- Where value concentrates, and at what price level.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing geographic — box position carries no map meaning.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- Only the top 25 areas are drawn.

VALIDATION PERFORMED

Box values reconcile with the Pareto chart in Insights.

4.3.4 Volume against price

Each bubble is an area.

In one sentence

A market map: busy areas to the right, expensive areas at the top.

WHAT IS THIS GRAPH?

A scatter of the 30 busiest areas: activity on one axis, median price on the other, bubble size for total value.

WHY IS THIS GRAPH USED?

Separates busy-and-cheap from quiet-and-expensive at a glance.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take the thirty busiest areas.
2. Plot each one by how many sales it had (across) and its middle sale price (up).
3. Size the bubble by the total money spent there, and colour it by price per square metre.

WORDS USED ON THIS CHART

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

Total value — Every sale price in the group added together — how much money changed hands.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	area_name_en, actual_worth, meter_sale_price

DATA PREPARATION

Top 30 areas by transaction count.

CALCULATION

```
X = count(rows); Y = median(actual_worth); size = sum(actual_worth); colour = median(meter_sale_price).
```

AXES AND LEGEND

Element	What it shows
X-axis	Number of transactions.
Y-axis	Median sale price (AED).
Legend	Colour = median rate per m ² . Bubble size = total value transacted.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Right = busy. Up = expensive. Big = a lot of money in total.
- The top-right corner is where both activity and price are high.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

A market map: which areas are liquid, which are premium, and which are both.

WHAT THIS GRAPH TELLS US

- How activity and price level relate across areas.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing causal — a busy area is not busy **because** it is priced the way it is.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- Only the 30 busiest areas appear.

VALIDATION PERFORMED

Values reconcile with the area detail table.

4.3.5 Locality zones

Median rate per m² by the zone label carried in the dataset.

In one sentence

What each broad kind of location costs — waterfront, central, suburban and so on.

WHAT IS THIS GRAPH?

The dataset's own six-way zone classification (Prime Central, Prime Waterfront, Urban Core, Suburban, Outer/Remote) against realised rate.

WHY IS THIS GRAPH USED?

A coarser, more communicable view than 69 individual areas.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Group every sale by the zone label the dataset carries.
2. For each zone, find the middle price per square metre.
3. Draw that as bars, with the number of sales as a line so you can see how much weight each zone carries.

WORDS USED ON THIS CHART

Locality zone — A six-way grouping of Dubai areas carried in the dataset — Prime Central, Prime Waterfront, Urban Core, Suburban, Outer/Remote and Unknown.

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	Locality Zone, meter_sale_price, actual_worth

DATA PREPARATION

Unknown zone excluded. Zones with fewer than 100 transactions dropped.

CALCULATION

Bars = median(meter_sale_price) per zone; line = count (rows).

AXES AND LEGEND

Element	What it shows
X-axis	Locality zone.
Y-axis	Median rate per m ² (AED).
Secondary (right) axis	Number of transactions.
Legend	Blue bars = price level. Dotted amber = how many deals sit behind it.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Compare bar heights for price, and always check the dotted line for how much data supports each bar.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

Waterfront, central, suburban — what each broad location type costs.

WHAT THIS GRAPH TELLS US

- The price gradient across Dubai's broad location types.

WHAT THIS GRAPH DOES NOT TELL US

- How the zone labels were assigned — that came with the dataset.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- Zone is a supplied label, not recomputed here.

VALIDATION PERFORMED

Zone medians were checked against the area-level ranking for consistency.

4.3.6 Nearest metro station

Median rate per m² grouped by each unit's nearest station.

In one sentence

Which metro corridors the expensive stock sits on.

WHAT IS THIS GRAPH?

Units grouped by the metro station recorded as nearest to them, ranked by median rate.

WHY IS THIS GRAPH USED?

Metro proximity is a standard value question in any city.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Group every sale by the metro station recorded as its nearest.
2. For each station, find the middle price per square metre.
3. Rank the stations.

WORDS USED ON THIS CHART

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	nearest_metro_en, meter_sale_price

DATA PREPARATION

Stations with fewer than 500 transactions are excluded; top 15 shown.

CALCULATION

median(meter_sale_price) per station.

AXES AND LEGEND

Element	What it shows
X-axis	Median rate per m ² (AED).
Y-axis	Nearest metro station.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- This ranks **neighbourhoods that those stations serve**, at least as much as it ranks the stations themselves.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

Which metro corridors the expensive stock sits on.

WHAT THIS GRAPH TELLS US

- Which metro corridors carry the most expensive stock.

WHAT THIS GRAPH DOES NOT TELL US

- It does **not** measure the value of being near a metro. That comparison is in the Price section, and it comes out negative — Dubai's most expensive waterfront stock is largely not on the metro network.
- Distance is not recorded, only which station is nearest.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- nearest_metro_en is blank for some units; those are excluded from this chart.

VALIDATION PERFORMED

Station groupings were checked against area names for plausibility.

4.4 Property

4.4.1 Layout mix

What buyers are actually buying.

In one sentence

What people are actually buying — studios, one-beds, two-beds.

WHAT IS THIS GRAPH?

Transaction counts for each bedroom configuration.

WHY IS THIS GRAPH USED?

Establishes what the typical Dubai unit sale actually is.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take every sale in your selection.
2. Count how many were studios, how many one-bedroom, and so on.
3. Draw one bar per layout.

WORDS USED ON THIS CHART

Layout — The bedroom configuration: studio, 1 B/R, 2 B/R and so on.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	rooms_en

DATA PREPARATION

Layouts ordered Studio → 1 B/R → ... → Penthouse.

CALCULATION

count (rows) per rooms_en value.

AXES AND LEGEND

Element	What it shows
X-axis	Layout.
Y-axis	Number of transactions.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Three layouts — 1 B/R, 2 B/R and Studio — make up the overwhelming majority of the market.

HOW TO EXPLAIN IT TO A CLIENT**In plain language**

Dubai's residential market is, overwhelmingly, a one- and two-bedroom apartment market.

WHAT THIS GRAPH TELLS US

- The shape of demand by unit type.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about supply or what is available unsold.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.

VALIDATION PERFORMED

Counts sum to the filtered row count; nine distinct layouts confirmed present with no nulls.

4.4.2 Typical size by layout

Median floor area for each layout.

In one sentence

How big a Dubai studio, one-bed or two-bed actually is.

WHAT IS THIS GRAPH?

The median recorded floor area of each bedroom configuration.

WHY IS THIS GRAPH USED?

A reference point when comparing listings, and the denominator behind every rate-per-m² figure on this dashboard.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Group the sales by layout.
2. For each layout, find the middle floor area.
3. Draw one bar per layout.

WORDS USED ON THIS CHART

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Layout — The bedroom configuration: studio, 1 B/R, 2 B/R and so on.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	rooms_en, procedure_area

DATA PREPARATION

Filtered rows only.

CALCULATION

median(procedure_area) per layout.

AXES AND LEGEND

Element	What it shows
X-axis	Layout.
Y-axis	Median unit size (m ²).

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Sizes step up cleanly: about 40 m² for a studio, 73 m² for a 1-bed, 120 m² for a 2-bed.

HOW TO EXPLAIN IT TO A CLIENT**In plain language**

What a studio, a one-bed and a two-bed actually measure in Dubai.

WHAT THIS GRAPH TELLS US

- What each layout label means in floor area.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about layout quality or usable space.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- `procedure_area` is the registered area, which may include balconies and shared allocations depending on the registration.

VALIDATION PERFORMED

Medians were checked to increase monotonically with bedroom count.

4.4.3 Rate per m² by layout

One panel per layout, so the distributions can be compared side by side.

In one sentence

How much a square metre costs in each size of apartment — and how widely that varies inside each one.

WHAT IS THIS GRAPH?

A small-multiples box plot: each layout gets its own panel showing the middle 50% of its transactions, its median, and its typical range.

WHY IS THIS GRAPH USED?

A single combined box plot compressed nine layouts on top of each other and was unreadable. Separate panels let each distribution be seen.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Group every sale by layout, and drop any layout with fewer than 100 sales (those are listed underneath rather than deleted).
2. For each layout, work out the price per square metre at the quarter, half and three-quarter points of its sales.
3. Draw a box from the quarter point to the three-quarter point, with a line at the middle — so the box holds the middle half of that layout's market.
4. Give each layout its own panel, all on the same scale, so they can be compared side by side.

WORDS USED ON THIS CHART

Quartile — Split the sales into four equal groups by price. The box in a box plot runs from the end of the first group to the end of the third — the middle half of the market.

Whisker — The line reaching out of a box. It stretches to the furthest sale still inside the normal range; anything past it is an unusual deal.

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

Layout — The bedroom configuration: studio, 1 B/R, 2 B/R and so on.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	rooms_en, meter_sale_price

DATA PREPARATION

Layouts with fewer than **100** transactions in the current selection are shown in a note rather than plotted, because a box drawn from a handful of sales is not meaningful. No outliers are deleted — the whiskers reach the furthest transaction within $1.5 \times \text{IQR}$ and anything beyond is summarised in the table below the chart.

CALCULATION

Per layout: 25th percentile, median, 75th percentile, and whiskers at the furthest observation inside $1.5 \times \text{IQR}$. Quartiles are computed in pandas rather than in the browser; the drawn statistics are identical.

AXES AND LEGEND

Element	What it shows
X-axis	Layout (one panel each).
Y-axis	Rate per m ² (AED). All panels share the same scale so they can be compared.
Legend	Each panel is coloured for identification only; the transaction count is printed under each panel.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- The box is the middle 50% of transactions; the line inside it is the median.
- The whiskers show the typical range; the table lists what falls beyond them.
- Compare median lines across panels to see how price per m² varies by layout.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

Bigger apartments in Dubai are not cheaper per square metre — they are more expensive per square metre. The value is in the location and the building, not in buying bulk.

WHAT THIS GRAPH TELLS US

- How rate per m² varies across layouts, and how wide the spread is within each.
- In this data, larger layouts command a **higher** median rate per m², not a lower one: about AED 14,845/m² for a studio against AED 19,943/m² for a 4-bed.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about total price — a studio at a high rate is still a cheaper home.
- Nothing about which layout is the better investment.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- Layouts with very few sales (6 B/R, 7 B/R) are excluded from the panels and reported in the note instead.

VALIDATION PERFORMED

Quartiles were recomputed directly in pandas and matched. The earlier claim that smaller units cost more per m² was tested against the data and **did not hold**, so it was removed.

4.4.4 Off-plan vs existing

Share of transactions by registration type.

In one sentence

How much of the market is buying from a developer before completion, versus buying a finished home.

WHAT IS THIS GRAPH?

The split between properties bought before completion (off-plan) and already-built stock (existing).

WHY IS THIS GRAPH USED?

It is the single biggest structural divide in the Dubai market and it affects almost every other figure on this dashboard.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take every sale and read its registration type.
2. Count how many are off-plan and how many are existing.
3. Show the two as shares of the whole.

WORDS USED ON THIS CHART

Off-plan — Bought from the developer before the building was finished.

Existing property — Already built and completed at the time it was sold. Also called resale or ready property.

Registration type — The Land Department's label for whether a sale was off-plan or existing. It is recorded on every transaction, so nothing here is guessed.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	reg_type_en

DATA PREPARATION

None. `reg_type_en` is a clean two-value field with no missing rows.

CALCULATION

`count ( rows )` per registration type, shown as shares.

AXES AND LEGEND

Element	What it shows
Legend	One slice per registration type.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- A majority off-plan share means a developer-led market.

HOW TO EXPLAIN IT TO A CLIENT**In plain language**

How much of Dubai's market is buying off-plan from a developer versus buying a finished home.

WHAT THIS GRAPH TELLS US

- Whether the market is being driven by new supply or by resale.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about completion risk or delivery timing.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.

VALIDATION PERFORMED

Verified against the raw registry: the same two values, `reg_type_id` 0 = Off-Plan and 1 = Existing, with no nulls.

4.4.5 Transaction procedure

The registration procedure recorded against each sale.

In one sentence

The legal form each sale took, as recorded by the Land Department.

WHAT IS THIS GRAPH?

The land-department procedure under which each sale was registered.

WHY IS THIS GRAPH USED?

Different procedures represent different kinds of transaction, and the mix shifts as the market shifts between new and resale stock.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take every sale and read its procedure name.
2. Count how many fall under each procedure.
3. Show them as shares.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	procedure_name_en

DATA PREPARATION

Top six procedures shown.

CALCULATION

`count ( rows )` per procedure.

AXES AND LEGEND

Element	What it shows
Legend	One slice per procedure.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- 'Sell – Pre registration' corresponds to off-plan; 'Sell' to completed property.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

The legal form each sale took — useful context, rarely the headline.

WHAT THIS GRAPH TELLS US

- The registration composition of the market.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about price.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- Only the six most common procedures are shown.

VALIDATION PERFORMED

Procedure counts cross-checked against the registration-type split.

4.4.6 Size against price

Each dot is one transaction.

In one sentence

Every sale as one dot: bigger homes cost more, but how much more depends heavily on where they are.

WHAT IS THIS GRAPH?

A scatter of unit size against sale price, coloured by layout.

WHY IS THIS GRAPH USED?

Shows the size-price relationship directly, and makes unusual deals visible.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take a large random sample of the sales in your selection (drawing every one would freeze the browser).
2. Plot each as a dot — floor area across, price up.
3. Colour the dots by layout.

WORDS USED ON THIS CHART

Layout — The bedroom configuration: studio, 1 B/R, 2 B/R and so on.

Distribution — The shape of the data — where most sales sit and how far the rest spread out.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	procedure_area, actual_worth, rooms_en

DATA PREPARATION

A deterministic random sample of 15,000 transactions (`random_state=42`) is drawn so the chart stays responsive. The sample size is printed on screen and the shape of the relationship is unchanged.

CALCULATION

No aggregation — raw transaction values.

AXES AND LEGEND

Element	What it shows
X-axis	Unit size (m ²).
Y-axis	Sale price (AED).
Legend	Colour = layout.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- The upward cloud is the basic size-price relationship.
- Points well above the cloud for their size are premium deals; well below are discounts or unusual registrations.

HOW TO EXPLAIN IT TO A CLIENT**In plain language**

Size explains a lot of price in Dubai — but far from all of it. The vertical spread at any given size is the location and building premium.

WHAT THIS GRAPH TELLS US

- How strongly size drives price, and how much scatter there is around it.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about **why** a specific point sits where it does.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- It is a sample, so the very rarest transactions may not appear.

VALIDATION PERFORMED

Sample medians were compared with full-population medians and agree.

4.5 Price

4.5.1 How prices are moving

Median sale price and median rate per m², monthly, with optional smoothing.

In one sentence

Are prices going up? Watch the teal line — that is price per square metre, which does not move just because people bought bigger homes this month.

WHAT IS THIS GRAPH?

Two monthly series: the median total price paid, and the median price per square metre. A control lets you show the actual monthly observations, a smoothed trend, or both together.

WHY IS THIS GRAPH USED?

These answer two different questions. Median total price moves when the **mix** of what people buy changes. Median rate per m² is the cleaner read on whether Dubai property itself is getting more expensive.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Put every sale into the month it happened in.
2. For each month, find the middle sale price and the middle price per square metre.
3. Draw those as two lines against two different scales.
4. For the smoothed view, replace each month with the middle value of itself and its two neighbours — the real monthly figures stay available and are listed in the table underneath.

WORDS USED ON THIS CHART

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

Rolling median — For each month, take that month together with the one before and the one after, and use the middle value of the three. It calms the zig-zag without moving the level or hiding anything — the real monthly figures are still there.

Partial year — A year the data does not cover in full. 2026 stops on 6 August, so eight months of it are being compared with somebody else's twelve.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	year_month, instance_date, actual_worth, meter_sale_price

DATA PREPARATION

Transactions grouped by calendar month. Every month in the series carries at least 814 transactions (median 2,437), so the month-to-month movement is genuine market and mix churn, not small-sample noise — smoothing is applied for readability only and never replaces the underlying observations.

CALCULATION

Per month: `median(actual_worth)` and `median(meter_sale_price)`. The smoothed series is a **centred 3-month rolling median** of those monthly medians. A rolling median is used rather than a mean because it is resistant to a single unusual month. Smoothing reduces the standard deviation of month-on-month change in the rate series from 7.6% to 4.4% without moving the level.

AXES AND LEGEND

Element	What it shows
X-axis	Month.
Y-axis	Median sale price (AED).
Secondary (right) axis	Median rate per m ² (AED/m ²).
Legend	Blue = median sale price. Teal = median rate per m ² . When both views are shown, the faint line is the actual monthly observation and the solid line is the smoothed trend.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Use **Smoothed trend** to read direction, and **Actual monthly** to see the real observations and any single unusual month.
- Blue and teal diverging means the mix of what is being bought is changing, not just the price level.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

“Are prices going up?” Watch the teal line: that is price per square metre, which is not affected by whether people happened to buy bigger homes this month.

WHAT THIS GRAPH TELLS US

- The direction and pace of price movement, on two different measures.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about individual properties or specific areas.
- The smoothed line is a reading aid, not a forecast.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- The final point (August 2026) covers only 6 days and 2,157 transactions against about 10,800 in a full month. It is flagged on the chart and is the cause of the sharp drop at the right-hand edge.
- A centred rolling window means the first and last month are computed from fewer observations than the middle of the series.

VALIDATION PERFORMED

Monthly medians were recomputed directly from the parquet and matched. Every month's transaction count was checked for thin samples — the smallest is 814 (May 2020) — confirming the jaggedness is real and not a data artefact.

4.5.2 Off-plan vs existing pricing

Median rate per m² by year, plus the off-plan premium or discount.

In one sentence

Off-plan property has sold for more per square metre than finished property in every year of this dataset. This shows by how much.

WHAT IS THIS GRAPH?

Two yearly lines — one for off-plan sales, one for existing property — and a companion chart showing the gap between them as a percentage.

WHY IS THIS GRAPH USED?

To answer directly whether off-plan trades at a premium or a discount, rather than leaving it to be eyeballed from two lines.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Split each year's sales into off-plan and existing.
2. Find the middle price per square metre for each side.
3. Skip any year with fewer than 100 sales on either side.
4. Work out the gap as a percentage of the existing-property figure — positive means off-plan cost more.

WORDS USED ON THIS CHART

Off-plan — Bought from the developer before the building was finished.

Existing property — Already built and completed at the time it was sold. Also called resale or ready property.

Registration type — The Land Department's label for whether a sale was off-plan or existing. It is recorded on every transaction, so nothing here is guessed.

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	year, reg_type_en, meter_sale_price

DATA PREPARATION

reg_type_en was validated on the raw registry: exactly two values (reg_type_id 0 = Off-Plan Properties, 1 = Existing Properties) with no missing rows, so no reclassification was needed. Years with fewer than 100 transactions on either side are excluded from the premium calculation.

CALCULATION

Per year and registration type: $\text{median}(\text{meter_sale_price})$. Premium (%) = $(\text{off-plan median} - \text{existing median}) \div \text{existing median} \times 100$. Positive means off-plan is more expensive.

AXES AND LEGEND

Element	What it shows
X-axis	Year.
Y-axis	Median rate per m ² (AED/m ²) — and, on the companion chart, premium (%).
Legend	Blue = existing property. Teal = off-plan. On the premium chart, teal bars are a premium and red bars a discount.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Where the teal line sits above the blue line, buyers are paying more per m² for unbuilt stock.
- The premium chart states the size of that gap directly, year by year.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

Buyers in Dubai consistently pay more per square metre for off-plan than for finished property — but that is because off-plan stock is newer and better located. In the same building the gap is gone. Quote this chart together with the one below it, never on its own.

WHAT THIS GRAPH TELLS US

- Off-plan has traded at a **premium in every single year** from 2010 to 2026 in this dataset, ranging from +4.4% (2011) to +70.9% (2019).
- The premium peaked around 2022 (+69.7%) and has narrowed since, to +25.6% in 2026 — existing stock has been catching up.

WHAT THIS GRAPH DOES NOT TELL US

- It does **not** mean an unbuilt apartment is worth more than the finished one next to it. Held inside a single building, the premium disappears — see **Why off-plan looks more expensive** directly below this chart.
- It does **not** mean off-plan is overpriced. Off-plan and existing stock are different products in different buildings: newer, higher-graded, in newer master developments, with payment spread over construction.
- Nothing about completion risk, delivery delay or handover quality.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- **The comparison is not like-for-like, and the difference is not small.** The two groups differ systematically in building, grade, price tier, area and developer. Value every building at its own median rate and off-plan buyers are transacting in buildings worth AED 19,445/m² against AED 12,486/m² for existing buyers — a 55.7% gap, which is almost the whole headline.
- Off-plan prices are registered at contract date; existing prices at sale — so the two series carry slightly different timing meanings.
- The registry records the headline contract price. An off-plan price paid over three years of construction is not the same money as a resale price paid at once, and that difference is invisible here.

VALIDATION PERFORMED

Recomputed independently in pandas for all 17 years, with transaction counts on both sides of every comparison. The premium is positive in 17 of 17 years. The classification was verified against the raw registry rather than assumed, and against the completion dates: 84% of off-plan sales occur before the building is finished (median 1.6 years before) while existing sales occur a median 2.9 years after.

4.5.3 How amenities relate to price — headline comparison

Median rate per m² for units with an amenity against those without.

In one sentence

The raw comparison — everything with a feature against everything without it. It is shown for transparency, not for quoting.

WHAT IS THIS GRAPH?

For each amenity flag, the straight difference in median rate per m² between units that carry the flag and units that do not.

WHY IS THIS GRAPH USED?

It is the first question every client asks. It is also the most easily misread number on the dashboard, which is why the like-for-like comparison sits directly beside it.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Split the sales into two piles: those recorded with the feature and those recorded without.
2. Find the middle price per square metre in each pile.
3. Express the difference as a percentage of the “without” pile.
4. Do that for all five features.

WORDS USED ON THIS CHART

Amenity flag — A yes/no column recording whether a feature was noted for that unit. A “no” means it was **not recorded**, which is not the same as “definitely absent”.

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

Composition effect — When a difference between two groups turns out to be caused by the groups containing different kinds of thing, rather than by what you were measuring.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	has_parking, swimming_pool, balcony, elevator, metro, meter_sale_price

DATA PREPARATION

All five flags are complete 0/1 integers with **no missing values**. Nothing is imputed and no missing value is treated as a “No”. Critically, a 0 means “not recorded as present”, which is not the same as “absent” — see the limitations. A group must have at least 50 transactions on each side to appear.

CALCULATION

Difference (%) = (median rate with the amenity – median rate without) ÷ median rate without × 100.

AXES AND LEGEND

Element	What it shows
X-axis	Median rate difference versus units without the amenity (%).
Y-axis	Amenity.
Legend	Teal = higher median rate with the amenity. Red = lower.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Read this as a **description of two groups of properties**, not as the value of a feature.
- Always read it together with the like-for-like chart beneath it — the two answer different questions.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

This chart compares two piles of properties. It does not price a feature. The chart below it does the fairer comparison.

WHAT THIS GRAPH TELLS US

- How the two groups of properties differ in realised price per m².

WHAT THIS GRAPH DOES NOT TELL US

- It does **not** say what an amenity is worth, and it does not support any causal statement. Parking at +102% does not mean parking doubles a price.
- The groups differ in area, layout, building age and — decisively — in whether the sale was off-plan or existing.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- **Only `has\_parking` exists in the raw registry.** The pool, balcony, elevator and metro flags are engineered fields that exist only in the cleaned dataset, so they cannot be re-derived from the raw file.
- A 0 conflates “this property does not have it” with “this was never recorded”. Recording is far more complete for existing property than for off-plan: a balcony is recorded for 88.8% of existing sales but only 32.9% of off-plan sales.
- Because off-plan trades at a much higher rate per m² than existing (AED 17,879 vs AED 11,264 median), any flag that is under-recorded off-plan will show a spurious negative headline.

VALIDATION PERFORMED

Every median was recomputed directly from the parquet and matched to one decimal place (see `tests/verify_dubai_numbers.py`). Group sizes, encodings and null counts were checked for all five flags.

4.5.4 How amenities relate to price — like-for-like

The same comparison, held within area, layout, year and registration type.

In one sentence

The honest version of the amenity question — and the numbers to use in front of a client.

WHAT IS THIS GRAPH?

The amenity comparison repeated inside narrow, comparable groups of properties, then averaged across those groups.

WHY IS THIS GRAPH USED?

The headline comparison is dominated by *which* properties carry each flag. Comparing only within a single area, layout, year and registration type removes most of that, leaving a far fairer estimate.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Sort every sale into a matched set: same area, same layout, same year, same off-plan status.
2. Keep only the sets that hold at least 30 sales both with and without the feature.
3. Inside each set, compare the two middle prices per square metre.
4. Average those set-by-set results, letting bigger sets count for more.

WORDS USED ON THIS CHART

Like-for-like — Comparing only properties that match on the things that matter — same area, same size of apartment, same year, same off-plan status — so the comparison is not secretly about something else.

Amenity flag — A yes/no column recording whether a feature was noted for that unit. A “no” means it was **not recorded**, which is not the same as “definitely absent”.

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

Composition effect — When a difference between two groups turns out to be caused by the groups containing different kinds of thing, rather than by what you were measuring.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	has_parking, swimming_pool, balcony, elevator, metro, area_name_en, rooms_en, year, reg_type_en, meter_sale_price

DATA PREPARATION

Transactions are grouped into area × layout × year × registration-type cells. A cell is used only when it contains at least **30** transactions on each side of the comparison.

CALCULATION

Within each cell, $(\text{median rate with} - \text{median rate without}) \div \text{median rate without} \times 100$. Cell results are averaged across cells, weighted by the number of transactions in each cell.

AXES AND LEGEND

Element	What it shows
X-axis	Like-for-like median rate difference (%).
Y-axis	Amenity.
Legend	Teal = higher, red = lower. The table beside it reports how many comparable groups and transactions each figure rests on.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- This is the number to quote. Compare it with the headline above to see how much of the raw gap was simply property mix.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

Once you compare like with like — same area, same size of apartment, same year, same off-plan status — most of the dramatic amenity differences disappear. That is the honest answer.

WHAT THIS GRAPH TELLS US

- Controlling for area, layout, year and registration type, the picture changes completely: parking +5.2%, pool +7.6%, elevator +6.1%, balcony -2.1%, near-metro -8.2%.
- Four of the five headline results were composition effects, not amenity effects.

WHAT THIS GRAPH DOES NOT TELL US

- Still not causal. Cells control for four characteristics, not for building quality, floor level, view, finish or developer.
- It cannot tell you what a specific building should charge.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- Requiring 30 transactions on both sides of every cell restricts the analysis to the busier areas and common layouts.
- Parking rests on the fewest comparable cells, because 95% of the cleaned dataset carries the flag.

VALIDATION PERFORMED

Computed at three levels of control and compared: no control, area × layout × year, and area × layout × year × registration type. The progressive collapse of every headline figure is itself the evidence that the headline was a composition effect.

4.5.5 Where the price points are

Transactions by sale-price band.

In one sentence

Where Dubai's money actually gets spent — and it is not at the top end.

WHAT IS THIS GRAPH?

The market split into seven price bands, with counts and shares in a table beside the chart.

WHY IS THIS GRAPH USED?

The quickest way to see which price points the Dubai market actually trades at.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take seven fixed price brackets, from under 500K to 10M and above.
2. Put every sale into exactly one bracket. Each bracket includes its lower limit and stops just short of the next, so nothing is counted twice and nothing is left out.
3. Count the sales in each bracket and work out its share of the total.
4. Draw the counts as bars, with the same numbers in the table beside them.

WORDS USED ON THIS CHART

Price band — A price bracket, such as AED 1M – 2M. Every sale falls in exactly one band, so the bands add up to the whole market.

Percentile — The 90th percentile is the value that 90% of sales come in below. The 1st and 99th are used here to trim off the few extreme deals at each end.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	actual_worth

DATA PREPARATION

Bands are left-closed and right-open ($< 500K$ means below 500,000; $500K - 1M$ means 500,000 up to but not including 1,000,000), so every transaction falls in exactly one band and no transaction is counted twice.

Validated on the raw registry: no null, zero or negative sale prices, and no duplicate transaction identifiers.

CALCULATION

count (rows) per band; share = band count ÷ total filtered transactions × 100.

AXES AND LEGEND

Element	What it shows
X-axis	Price band (AED).

Element	What it shows
Y-axis	Number of transactions.
Legend	Bar labels show each band's share of the current selection.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- The tallest bars are where the market actually is.
- The right-hand tail is the luxury segment.
- **Check the note under the chart:** the sale-price slider defaults to the 1st–99th percentile, which truncates the top bands. Widen it to see them.

HOW TO EXPLAIN IT TO A CLIENT**In plain language**

Where Dubai's money actually gets spent — and it is not at the top end.

WHAT THIS GRAPH TELLS US

- Two-thirds of Dubai's residential unit sales fall between AED 500K and 2M.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about size or value for money — a 2M studio and a 2M three-bed are in the same band.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- The bands are fixed business bands, not derived from the distribution.
- The default price filter hides the top of the range. Unfiltered, the cleaned dataset contains 5,308 sales at AED 10M or more (0.6%).

VALIDATION PERFORMED

Recomputed on the raw registry, the raw residential-unit subset and the cleaned dataset. In all three the band counts sum exactly to the row count with zero unassigned rows, confirming the bands are exhaustive and mutually exclusive. Chart and table are generated from the same computed frame, so they cannot disagree.

4.5.6 Published price forecast

Quarterly ARIMA forecasts of rate per m², produced by the existing modelling work.

In one sentence

The forward view our modelling work already produced for this area, shown with its own error record so you can judge how far to trust it.

WHAT IS THIS GRAPH?

The fitted history and forward forecast of quarterly rate per m² for one area, with a confidence band, plus the model's published accuracy.

WHY IS THIS GRAPH USED?

To put a forward view next to the historical analysis, using the results the project's modelling pipeline already produced.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Read the forecast files the modelling pipeline saved earlier. Nothing is recalculated here.
2. Draw the model's fit against what actually happened, so you can see how well it tracked history.
3. Draw the forward quarters, with a shaded band for the range the model expects.
4. Show the model's published error score next to it.

WORDS USED ON THIS CHART

Arima — A standard statistical forecasting model. It learns the pattern of a series from its own history and projects it forward. It uses no outside information.

Confidence band — The range the model expects the true value to fall inside. A wide band means the model is not very sure.

Maape — Mean absolute percentage error — on average, how far the model's past predictions were from what actually happened, in percent. Lower is better.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

DATA SOURCE

Field	Value
Source	DERIVED
Dataset file	regions/dubai/*.csv (forecast artefacts)
What that file is	Outputs already produced by the project's Dubai modelling pipeline and stored with the repository. Nothing is refitted at run time.
Source columns	ds, area_name_en, yhat, actual, type, yhat_lower, yhat_upper, forecast_quarter, forecast_price, growth_factor, Test_MAPE, Test_MAE

DATA PREPARATION

Read directly from the stored CSV outputs. **Nothing is refitted at run time.** Only the modelled subset of areas is available, not all 69.

CALCULATION

None performed here — the ARIMA/SARIMA fitting was done by the existing pipeline. This chart displays its published output.

AXES AND LEGEND

Element	What it shows
X-axis	Quarter.
Y-axis	Rate per m ² (AED/m ²).
Legend	Grey = actual quarterly rate. Dotted teal = model fit to history. Blue = forecast. Shaded band = the forecast's uncertainty range.

FILTERS

The Dubai sidebar filters do not apply to this chart — it displays stored model output for a single area chosen from its own selector.

HOW TO READ IT

- Check the dotted fit against the grey actuals first: if the model tracked history poorly, treat the forecast with caution.
- A wider shaded band means less confidence.
- Read the Test MAPE beside the chart — it is the model's average error on data it had not seen.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

Our modelling team's forward view for this area, shown with its own error record so you can judge how much to trust it.

WHAT THIS GRAPH TELLS US

- The modelled forward path for the areas that were modelled.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about areas that were not modelled.
- A forecast is not a commitment; the band is part of the answer, not decoration.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- Covers only the subset of areas the pipeline modelled.
- Produced at a fixed point in time; it does not update with the dashboard data.
- Sidebar filters have no effect on it.

VALIDATION PERFORMED

Files are read as published; the app reports any that are missing rather than silently drawing an empty chart.

4.5.7 Why the raw amenity number is so much bigger

The same amenity, measured under progressively fairer comparisons.

In one sentence

Watch what happens to the big number as the comparison gets fairer. Reading down, each bar compares properties that match on one more thing — by the bottom bar you are comparing genuinely similar homes, and the answer can shrink, or flip.

WHAT IS THIS GRAPH?

One bar per comparison, read top to bottom. The top bar compares every unit carrying the amenity flag against every unit without it, whatever kind of property they are. Each bar below it compares only properties that also match on one more characteristic — first area, then layout, then year, then off-plan status. The bottom bar is the fair comparison.

WHY IS THIS GRAPH USED?

Because the raw amenity number is the single most misread figure on this dashboard, and telling somebody “it is a composition effect” does not help them. Watching the gap melt as the comparison gets fairer does.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Start with the raw comparison: every unit with the feature against every unit without it.
2. Now only compare units in the same area as each other. Recalculate.
3. Now also require the same layout. Then the same year. Then the same off-plan status.
4. At each step, work out the difference inside every matched set separately, then average the sets — bigger sets count for more.
5. A set needs at least 30 sales on each side to be used at all.

WORDS USED ON THIS CHART

Like-for-like — Comparing only properties that match on the things that matter — same area, same size of apartment, same year, same off-plan status — so the comparison is not secretly about something else.

Composition effect — When a difference between two groups turns out to be caused by the groups containing different kinds of thing, rather than by what you were measuring.

Amenity flag — A yes/no column recording whether a feature was noted for that unit. A “no” means it was **not recorded**, which is not the same as “definitely absent”.

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

Off-plan — Bought from the developer before the building was finished.

Existing property — Already built and completed at the time it was sold. Also called resale or ready property.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	has_parking, swimming_pool, balcony, elevator, metro, meter_sale_price, area_name_en, rooms_en, year, reg_type_en

DATA PREPARATION

Rows are filtered by the sidebar selection. For every level except the first, properties are grouped into matched sets and a set must hold at least 30 transactions on each side to be used, so a handful of deals cannot swing the answer.

CALCULATION

Level 0 is $(\text{median rate with} \div \text{median rate without} - 1) \times 100$ across the whole selection. Each level after that computes that same percentage **inside every matched set separately**, then averages the sets weighted by how many transactions each holds.

AXES AND LEGEND

Element	What it shows
X-axis	Difference in median rate per m ² , in percent.
Y-axis	How tightly the comparison is controlled.

Element	What it shows
Legend	Amber = the raw, uncontrolled comparison. Grey = partly controlled. Teal or red = the fair comparison, and its direction.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Start at the **top** bar — that is the raw number, the one people quote, and it is usually the biggest.
- Work downwards. Each bar removes one more reason the two groups of properties might differ for reasons that have nothing to do with the amenity.
- The **bottom bar is the answer**. If it is much smaller than the top one, the raw number was mostly describing which kind of property carries the label.
- The middle bars can move up as well as down. That is expected — each level is measured on a different set of comparable properties, so the sample changes as well as the control. Only the top and bottom bars are meant to be compared directly.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

“The big number compares two different kinds of property. When we compare like with like, the difference is this much.” That is the whole story in one picture.

WHAT THIS GRAPH TELLS US

- How much of an amenity's apparent price gap survives once you compare comparable properties.
- Which characteristic does the damage — the level where the bar suddenly shrinks is the one that was really driving the raw number.

WHAT THIS GRAPH DOES NOT TELL US

- It still does not prove causation. Even the bottom bar is an association measured across four controls, not an experiment.
- It cannot control for floor level, view, finish, service charge or developer — none of those are in the dataset.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- Tightening the comparison also shrinks the sample: the fairest level uses fewer transactions than the raw one. Both counts are on the hover.
- If the selection is narrow, some levels may not have any matched set large enough and will be missing from the chart.

VALIDATION PERFORMED

Every level was recomputed independently with plain pandas in `tests/verify_dubai_changes.py`, including the weighting and the 30-transaction floor.

4.5.8 Why off-plan looks more expensive

The off-plan premium, measured under progressively fairer comparisons.

In one sentence

If a finished flat you can walk through should be worth more than a drawing — and it should — this is the chart that explains the headline. The premium belongs to the buildings, not to being unbuilt.

WHAT IS THIS GRAPH?

One bar per comparison, read top to bottom. The top bar is the straight off-plan-against-existing comparison. Each bar below it compares only sales that also match on one more thing — same year, then same area, then same master development, then same project, then the same building. The bottom bar compares units in a single building in a single year.

WHY IS THIS GRAPH USED?

Because the headline contradicts common sense, and the contradiction is the interesting part. A finished apartment can be walked through, inspected and moved into; an off-plan one cannot. It should not be worth less. The ladder shows that it is not — the premium belongs to the buildings off-plan is sold in, not to being unbuilt.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Start with the straight comparison: every off-plan sale against every existing sale.
2. Now compare only sales from the same year. Recalculate.
3. Now only sales in the same area. Then the same master development. Then the same project. Then the same individual building.
4. At each step, work out the difference inside every matched set separately, then average the sets — bigger sets count for more.
5. A set needs at least 30 sales on each side to be used at all, so a handful of deals in one tower cannot decide the answer.

WORDS USED ON THIS CHART

Off-plan — Bought from the developer before the building was finished.

Existing property — Already built and completed at the time it was sold. Also called resale or ready property.

Registration type — The Land Department's label for whether a sale was off-plan or existing. It is recorded on every transaction, so nothing here is guessed.

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

Like-for-like — Comparing only properties that match on the things that matter — same area, same size of apartment, same year, same off-plan status — so the comparison is not secretly about something else.

Composition effect — When a difference between two groups turns out to be caused by the groups containing different kinds of thing, rather than by what you were measuring.

DATA SOURCE

Field	Value
Source	CLEANED

Field	Value
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	reg_type_en, meter_sale_price, area_name_en, master_project_en, project_name_en, building_name_en, year

DATA PREPARATION

Rows are filtered by the sidebar selection. For every level except the first, sales are grouped into matched sets and a set must hold at least 30 transactions on each side to be used.

CALCULATION

Level 0 is $(\text{off-plan median rate} \div \text{existing median rate} - 1) \times 100$ across the whole selection. Each level below computes that same percentage **inside every matched set separately**, then averages the sets weighted by how many transactions each holds. The same function computes the amenity ladder, so the two arguments are made the same way.

AXES AND LEGEND

Element	What it shows
X-axis	Difference in median rate per m ² between off-plan and existing, in percent.
Y-axis	How tightly the comparison is controlled.
Legend	Amber = the raw, uncontrolled comparison. Grey = partly controlled. Teal or red = the same-building comparison, and its direction.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- The **top** bar is the number on the chart above it — around +59% across the whole dataset.
- Area and master development barely dent it, because a single area contains both old cheap towers and new luxury ones.
- It collapses at **project** and **building** level. Inside one building, off-plan is not dearer — it is slightly cheaper.
- That is the answer: the premium is a statement about **which buildings** are sold off-plan, not about buying before completion.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

“Off-plan sells for more per square metre because off-plan stock is newer and better located — not because buyers pay a premium for something they cannot see. Put an off-plan and a finished unit in the same building and the gap disappears.”

WHAT THIS GRAPH TELLS US

- That the off-plan premium is a stock-quality effect, not a payment-for-risk effect.
- Roughly where the effect lives — between master development and individual building, i.e. in the specific tower rather than the neighbourhood.

WHAT THIS GRAPH DOES NOT TELL US

- It does not say a specific off-plan purchase is good or bad value. That depends on the building, the payment plan and the handover date.
- It cannot see payment terms. An off-plan headline price paid over three years is not the same money as a resale price paid at once, and the registry records only the headline.
- It says nothing about what happens to an off-plan unit's value after handover.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- The same-building level is a much smaller sample than the headline: a building needs 30 off-plan **and** 30 existing sales in the same year to qualify. Counts are on the hover, and a larger pooled-across-years figure is shown beside the chart as a cross-check.
- Buildings with both kinds of sale in one year are typically at or just past handover, so the same-building comparison leans towards recently completed stock rather than the whole market.
- Off-plan and existing are not identical products even inside one building — different floors, views and finishes are not in this dataset.

VALIDATION PERFORMED

Every level was recomputed independently with plain pandas in `tests/verify_dubai_changes.py`. The classification itself was checked against the completion dates: 84% of off-plan sales occur before the building is finished, a median 1.6 years before, while existing sales occur a median 2.9 years after — so the label means what it says.

4.6 Distribution

4.6.1 Sale price distribution

How sale prices are spread.

In one sentence

What a typical Dubai sale costs, and how unusual the very big ones are.

WHAT IS THIS GRAPH?

A histogram of sale prices across the current selection.

WHY IS THIS GRAPH USED?

Averages hide shape. This shows where transactions actually cluster.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take every sale price in your selection.
2. Trim the most extreme half-percent at each end so the chart is readable.
3. Slice the remaining range into 60 equal buckets.
4. Count how many sales land in each bucket and draw that as a bar.

WORDS USED ON THIS CHART

Histogram — Sales sorted into equal-width buckets; the height of each bar is how many landed in that bucket.

Distribution — The shape of the data — where most sales sit and how far the rest spread out.

Percentile — The 90th percentile is the value that 90% of sales come in below. The 1st and 99th are used here to trim off the few extreme deals at each end.

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	actual_worth

DATA PREPARATION

The chart trims the extreme 0.5% at each end so the bars remain legible; the dashed median line is computed on the **untrimmed** data. Binning is done in numpy rather than in the browser — the chart is identical, just far lighter.

CALCULATION

60 equal-width bins between the 0.5th and 99.5th percentiles; `count ( rows )` per bin.

AXES AND LEGEND

Element	What it shows
X-axis	Sale price (AED).
Y-axis	Number of transactions.
Legend	Dashed amber line marks the median.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- The long right tail is normal for property — a few very large deals pull the average above the median.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

What a typical Dubai transaction costs, and how unusual the big ones are.

WHAT THIS GRAPH TELLS US

- The shape of the market, not just its centre.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about size, location or quality.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- The visible range is trimmed at each end.

VALIDATION PERFORMED

Bin counts were checked to sum to the trimmed row count.

4.6.2 Rate per m² distribution

How rate per m² is spread.

In one sentence

Is Dubai one market or several? If there is more than one hump, there is more than one market.

WHAT IS THIS GRAPH?

A histogram of price per square metre.

WHY IS THIS GRAPH USED?

More than one peak would indicate genuinely separate market tiers rather than one continuous market.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take the price per square metre of every sale.
2. Trim the extremes at each end for readability.
3. Slice the range into 60 equal buckets and count what lands in each.

WORDS USED ON THIS CHART

Histogram — Sales sorted into equal-width buckets; the height of each bar is how many landed in that bucket.

Distribution — The shape of the data — where most sales sit and how far the rest spread out.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	meter_sale_price

DATA PREPARATION

Same as the sale-price histogram: 0.5% trimmed each end for display, median computed untrimmed.

CALCULATION

60 equal-width bins; count (rows) per bin.

AXES AND LEGEND

Element	What it shows
X-axis	Rate per m ² (AED/m ²).
Y-axis	Number of transactions.
Legend	Dashed amber line marks the median.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Look for whether the shape is one hump or several.

HOW TO EXPLAIN IT TO A CLIENT**In plain language**

Is Dubai one market or several? This is where you can see it.

WHAT THIS GRAPH TELLS US

- Whether Dubai prices as one market or as distinct tiers.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about which properties sit where in the distribution.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- The visible range is trimmed at each end.

VALIDATION PERFORMED

Bin counts sum to the trimmed row count.

4.6.3 Unit size distribution

How unit sizes are spread.

In one sentence

The standard sizes Dubai builds and sells.

WHAT IS THIS GRAPH?

A histogram of registered floor area.

WHY IS THIS GRAPH USED?

Clusters correspond to standard studio, one-bed and two-bed layouts.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take the floor area of every sale.
2. Trim the extremes at each end.
3. Slice the range into 60 equal buckets and count what lands in each.

WORDS USED ON THIS CHART

Histogram — Sales sorted into equal-width buckets; the height of each bar is how many landed in that bucket.

Distribution — The shape of the data — where most sales sit and how far the rest spread out.

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	procedure_area

DATA PREPARATION

0.5% trimmed each end for display.

CALCULATION

60 equal-width bins; count (rows) per bin.

AXES AND LEGEND

Element	What it shows
X-axis	Unit size (m ²).
Y-axis	Number of transactions.
Legend	Dashed amber line marks the median.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- The peaks line up with the standard layouts in the Property section.

HOW TO EXPLAIN IT TO A CLIENT**In plain language**

The standard sizes Dubai builds and sells.

WHAT THIS GRAPH TELLS US

- The physical shape of Dubai's residential stock as traded.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about layout quality or usable versus registered area.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.

VALIDATION PERFORMED

Peaks were cross-checked against median sizes per layout.

4.6.4 Sale price by registration type

Price spread for off-plan and existing property.

In one sentence

Off-plan and finished property are not just priced differently — they are spread differently.

WHAT IS THIS GRAPH?

Box plots of sale price for each registration type, on a log scale.

WHY IS THIS GRAPH USED?

Compares not just the typical price but the whole spread of each segment.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Split the sales into off-plan and existing.
2. For each side, find the quarter, half and three-quarter points of the price range.
3. Draw a box covering the middle half, with a line at the middle.
4. Use a multiplying scale on the price axis, because the values run from a few hundred thousand to hundreds of millions.

WORDS USED ON THIS CHART

Quartile — Split the sales into four equal groups by price. The box in a box plot runs from the end of the first group to the end of the third — the middle half of the market.

Whisker — The line reaching out of a box. It stretches to the furthest sale still inside the normal range; anything past it is an unusual deal.

Log scale — An axis where each step multiplies instead of adding (1M, 10M, 100M). Useful when the values run from small to enormous.

Off-plan — Bought from the developer before the building was finished.

Existing property — Already built and completed at the time it was sold. Also called resale or ready property.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.

Field	Value
Source columns	reg_type_en, actual_worth

DATA PREPARATION

Quartiles computed in pandas; whiskers reach the furthest observation within $1.5 \times \text{IQR}$. No transaction is deleted.

CALCULATION

25th percentile, median, 75th percentile and $1.5 \times \text{IQR}$ whiskers per registration type.

AXES AND LEGEND

Element	What it shows
X-axis	Registration type.
Y-axis	Sale price (AED) on a logarithmic scale, so the full range fits.
Legend	One box per registration type.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- The box is the middle 50%. Note the **log scale** — equal distances represent equal ratios, not equal amounts.

HOW TO EXPLAIN IT TO A CLIENT**In plain language**

Off-plan and existing are not just priced differently — they have different spreads.

WHAT THIS GRAPH TELLS US

- How the two segments differ in both level and spread.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about rate per m^2 — see the Price section for that.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- A log axis makes differences look smaller than they are.

VALIDATION PERFORMED

Quartiles recomputed in pandas and matched.

4.6.5 How the price distribution has changed

Rate per m^2 , year by year.

In one sentence

The market has not only moved up — it has spread out. That matters when you price one specific unit.

WHAT IS THIS GRAPH?

One violin per year showing the full distribution of rate per m².

WHY IS THIS GRAPH USED?

A median only tells you the centre. This shows whether the market is also spreading out.

WHAT HAPPENS INSIDE THIS CHART, STEP BY STEP

1. Take a large random sample of the sales, so the chart stays responsive.
2. Group them by year.
3. For each year, draw a smoothed outline of where its prices per square metre sit — wider means more sales at that level.
4. Put a small box plot inside each outline for the middle half.

WORDS USED ON THIS CHART

Violin — A smoothed outline of the distribution. Where it bulges, there are more sales at that price.

Distribution — The shape of the data — where most sales sit and how far the rest spread out.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

Quartile — Split the sales into four equal groups by price. The box in a box plot runs from the end of the first group to the end of the third — the middle half of the market.

DATA SOURCE

Field	Value
Source	CLEANED
Dataset file	data/dubai/latest_combined_data.parquet
What that file is	818,838 residential unit sales registered in Dubai, 2010 – Aug 2026, enriched by the project with time parts, unit attributes, amenity flags and building / developer scoring.
Source columns	year, meter_sale_price

DATA PREPARATION

A deterministic random sample of 45,000 transactions (`random_state=42`) keeps the chart responsive; the distribution shape is preserved.

CALCULATION

Kernel density estimate of `meter_sale_price` per year, with an inner box plot.

AXES AND LEGEND

Element	What it shows
X-axis	Year.
Y-axis	Rate per m ² (AED/m ²).
Legend	One violin per year; colour is for identification only.

FILTERS

All seven Dubai sidebar filters apply: year, locality zone, area, layout, registration type, sale-price range and unit-size range.

HOW TO READ IT

- Wider at a given height = more transactions at that price level.
- Watch two things: the centre moving up, and the shape getting wider. The second means the market is spreading, not just rising.

HOW TO EXPLAIN IT TO A CLIENT

In plain language

The market has not just moved up — it has spread out. That matters for how you price a specific unit.

WHAT THIS GRAPH TELLS US

- How the whole price distribution has evolved, not just its midpoint.

WHAT THIS GRAPH DOES NOT TELL US

- Nothing about which segments drove the change.

LIMITATIONS

- Covers registered residential **unit** (apartment) sales only — villas, land and whole-building transactions are not in this dataset.
- 2026 is a partial year: the data ends **6 August 2026**, so 2026 totals are not comparable with a full year.
- The sale-price and unit-size sliders start at the 1st–99th percentile, so the most extreme deals are excluded until you widen them.
- Based on a sample.
- 2026 covers eight months only.

VALIDATION PERFORMED

Sampled medians per year were compared with full-population medians and agree.

Appendix A. The amenity results, explained

This appendix exists because the amenity chart is the easiest number on the dashboard to misread, and the most likely to be quoted out of context.

A.1 The headline figures, and what happens when you control for property mix

The first column is the straight comparison across the whole dataset. The next two repeat it inside progressively narrower groups of comparable properties.

Amenity	Units with	Units without	Headline	Same area, layout & year	...and same registration type
Parking	779,264	39,574	+117.1%	+21.1%	+5.2%
Swimming pool	237,348	581,490	-6.7%	-0.7%	+7.6%
Balcony	466,003	352,835	-30.4%	-15.6%	-2.1%
Elevator	313,907	504,931	-13.7%	-3.5%	+6.1%
Near a metro station	590,572	228,266	-10.0%	-12.1%	-8.2%

The single most important line in this document

Four of the five headline figures reverse or collapse once you compare like with like. The right-hand column is the one to quote. The headline column describes two different groups of properties, not the value of a feature.

A.2 Why the headline figures are wrong — the recording effect

Building attributes are recorded far more completely for finished property than for off-plan sales. Because off-plan trades at a much higher rate per square metre, any flag that is under-recorded on off-plan sales produces a false negative headline.

Registration type	Balcony recorded (%)	Not recorded (%)	Median rate (AED/m ²)	Share of deals
Existing Properties	88.8%	11.2%	11,264	42.9%
Off-Plan Properties	32.9%	67.1%	17,879	57.1%

Read that table slowly. A balcony is recorded for the large majority of existing sales and a minority of off-plan sales. Off-plan sells for far more per square metre. So "no balcony recorded" is largely a proxy for "off-plan" — and the -30% headline is measuring registration type, not balconies.

The proof: the balcony gap reverses inside each registration type

Registration type	Median rate WITH balcony	Median rate WITHOUT	Deals with	Deals without	Difference
Existing Properties	11,440	9,818	312,304	39,323	+16.5%
Off-Plan Properties	15,228	19,148	153,699	313,512	-20.5%

What a sign flip means

Within existing property, balconies are associated with a higher rate. Within off-plan property, with a lower

one. When a comparison reverses depending on how you slice the data, the overall figure is describing the mix of properties, not the feature. This is a textbook composition effect, and it is why the like-for-like number is the one to use.

A.3 Parking: what “+102%” actually means

The correct statement is: **“properties recorded with parking show a higher observed median rate per square metre than properties without, in this dataset.”** It is not “parking doubles the price”. Here is the evidence for why not.

On the raw registry, the flag only ever applies to apartments

Property type	With parking	Without parking
Building	0	16,083
Land	0	109,483
Unit	970,683	68,007
Villa	0	185,597

Every villa, land and whole-building row carries a zero. On the raw file the comparison is therefore largely apartments against land and villas — which is not a parking comparison at all.

Restricting to residential apartments only, on the raw registry

	With parking	Without parking
Transactions	870,910	57,579
Median rate (AED/m ²)	15,227	6,711
Median size (m ²)	79	50
Difference	+126.9%	

What the “no parking” apartments actually are

- Overwhelmingly studios: 53.3% of the no-parking group against 21.6% of the with-parking group.
- Overwhelmingly resale rather than new: 89.8% existing property against 40.6%.
- Heavily concentrated in one affordable area — Al Warsan First alone accounts for 56.2% of them.
- Much smaller: a median of 50 m² against 79 m².

The practical explanation

Parking is acting as a marker for a different kind of property: a larger, newer, off-plan apartment in a mid-market or premium development, rather than a small resale studio in an affordable area. Compare like with like — same area, same layout, same year, same registration type — and the gap falls from +117.1% to +5.2%, measured across 115 comparable groups covering 73,891 transactions.

Parking is genuinely associated with a modest price premium. It is not associated with a doubling.

A.4 Balcony: why a negative number does not mean “balconies are bad”

The correct statement is: **“properties recorded with a balcony show a lower observed median rate per square metre than those without, in this dataset.”** It is not “balconies reduce value”.

- The headline gap is -30.4%. Held within the same area, layout and year it narrows to -15.6%. Adding registration type to the control brings it to -2.1% — effectively nothing.
- Registration type is the driver. Balconies are recorded for most existing-property sales and a minority of off-plan sales, and off-plan sells for far more per square metre.
- Within existing property alone, the balcony association is clearly positive — see A.2.
- Nothing in this dataset supports a claim that a balcony reduces what a buyer will pay.

How to answer the client question

“Does a balcony hurt the price?” No. What the raw chart shows is that the kind of property recorded as having a balcony — typically completed resale stock — sells for less per square metre than off-plan stock does. Compare two similar apartments in the same building type and the balcony effect is close to zero, and positive within the resale market.

A.5 Rules for quoting these numbers

- Quote the like-for-like figure, not the headline.
- Say “properties with X show...”, never “X increases the price by...”.
- State the sample: how many comparable groups and how many transactions.
- Remember that a zero in these fields means “not recorded as present”, which is not the same as “confirmed absent”.
- Only the parking flag can be checked against the raw registry; the other four exist only in the cleaned dataset.

Appendix B. Year-over-year validation

Each year is compared with the immediately preceding year — never against a fixed baseline. The table below is the full working.

Year	Transactions	Volume vs prior yr	Median price (AED)	Median rate (AED/m ²)	Rate vs prior yr
2010	24,435	base	904,800	9,727	base
2011	18,355	-24.9%	950,000	9,334	-4.0%
2012	21,771	+18.6%	935,000	9,832	+5.3%
2013	38,942	+78.9%	990,590	10,905	+10.9%
2014	34,432	-11.6%	988,950	12,206	+11.9%
2015	25,413	-26.2%	918,860	11,355	-7.0%
2016	25,328	-0.3%	889,409	11,245	-1.0%
2017	27,811	+9.8%	922,888	11,956	+6.3%
2018	19,348	-30.4%	850,000	11,840	-1.0%
2019	21,398	+10.6%	990,140	12,157	+2.7%
2020	18,405	-14.0%	872,828	10,962	-9.8%
2021	33,102	+79.9%	1,050,000	12,151	+10.8%
2022	58,164	+75.7%	1,292,888	15,928	+31.1%
2023	88,537	+52.2%	1,265,000	16,117	+1.2%
2024	128,768	+45.4%	1,280,000	17,373	+7.8%
2025	159,477	+23.8%	1,300,000	18,370	+5.7%
2026	75,152	-52.9%	1,250,000	18,359	-0.1%

Years present: 2010–2026, with no gaps. Negative volume years: 2011, 2014, 2015, 2016, 2018, 2020, 2026.
Negative rate years: 2011, 2015, 2016, 2018, 2020, 2026.

2026 is a partial year — read its bar with care

The data ends 2026-08-06, so 2026 covers 8 months. Comparing 8 months against a full 12 produces -52.9%, which is an artefact of the calendar, not a market collapse.

Like for like — the same 8 months of each year — volume is -23.8% (75,152 transactions against 98,637) and the median rate is +1.13%.

Medians are far less sensitive to a shorter year than counts are, which is why the rate line is broadly reliable even for the partial year while the volume bar is not.

The negative years were each traced back to their own transaction count and median before being accepted. They are genuine market contractions consistent with the known Dubai cycle, and they have been kept. Negative results are not removed from this dashboard for being unwelcome.

Appendix C. Off-plan premium validation

Classification was validated on the raw registry before any calculation: Existing Properties (1,123,176), Off-Plan Properties (639,086), with 0 missing values. Two clean values, so no reclassification was needed and none was invented.

Year	Off-plan median	Existing median	Off-plan deals	Existing deals	Premium
2010	10,815	9,082	8,488	15,947	+19.1%
2011	9,689	9,281	1,811	16,544	+4.4%
2012	12,279	9,610	2,604	19,167	+27.8%
2013	15,134	10,408	6,856	32,086	+45.4%
2014	14,447	11,164	9,683	24,749	+29.4%
2015	12,443	10,766	9,847	15,566	+15.6%
2016	11,698	10,798	13,060	12,268	+8.3%
2017	12,920	11,064	16,118	11,693	+16.8%
2018	13,970	10,110	10,797	8,551	+38.2%
2019	15,457	9,046	12,586	8,812	+70.9%
2020	14,400	8,614	9,294	9,111	+67.2%
2021	15,613	10,411	15,083	18,019	+50.0%
2022	19,421	11,443	32,034	26,130	+69.7%
2023	19,394	12,110	54,381	34,156	+60.1%
2024	19,048	13,066	89,164	39,604	+45.8%
2025	19,747	14,383	118,070	41,407	+37.3%
2026	18,918	15,064	57,335	17,817	+25.6%

Off-plan traded at a premium in 17 of 17 years. That figure is correct. What it means is a different question, and the rest of this appendix answers it.

C.1 Does the label mean what it says?

Yes. On the 50.8% of rows that carry a completion date, **84.3%** of off-plan sales happen **before** the building is finished — a median of **1.6 years** before. Existing sales happen a median of **2.9 years** **after** completion. The registration type is not a proxy for something else; off-plan really is unbuilt property.

Registration type	Rows with a completion date	Median years vs completion	Sold before completion
Off-Plan Properties	187,026	-1.6	84.3%
Existing Properties	228,545	+2.9	2.2%

C.2 Then why is unbuilt property dearer than finished property?

It is not. The two labels are attached to different buildings. Tightening the comparison one step at a time — the same method used for the amenity figures — the premium survives area and master development, then collapses at project and building level.

Comparison	Premium	Matched groups	Transactions
Everything against everything	+58.7%	1	818,838
Same year	+40.7%	17	818,838
Same area and year	+37.3%	402	710,314
Same master development and year	+31.7%	407	631,901
Same project and year	-8.3%	106	42,841
Same building and year	-1.5%	83	14,911

The line to remember

Across the whole dataset off-plan looks +59% dearer per square metre. Inside a single building in a single year it is -1.5% — and pooled across all years over 896 buildings and 298,269 transactions (36.4% of the dataset) it is -7.5%, with off-plan dearer in only 30% of those buildings.

C.3 The stock test

Value every building at its own median rate, then ask what the average building is worth on each side of the split. Off-plan buyers transact in buildings worth **AED 19,468/m²**; existing buyers in buildings worth **AED 12,472/m²** — a gap of **+56.1%**, measured across 3,764 buildings. The headline premium is +58.7%. The difference in stock alone accounts for effectively all of it, leaving nothing that needs to be explained by off-plan status.

C.4 The two sides are not the same product

Share of each side's transactions, by the price tier the dataset carries:

Price tier	Off-plan	Existing	Difference
Affordable ○	4%	29%	-25 pp
Luxury ★	29.8%	13%	+16.8 pp
Premium ▲	39.2%	28.6%	+10.6 pp
Mid-Range ●	18%	27.1%	-9.1 pp
Ultra-Luxury ◆	7.9%	2.3%	+5.6 pp

And by building grade:

Building grade	Off-plan	Existing	Difference
C	16.6%	37.1%	-20.4 pp
B	29.6%	16.8%	+12.9 pp
A	23.9%	13.3%	+10.6 pp
D	1.4%	7.4%	-6 pp
A+	7.1%	2.2%	+4.9 pp

C.5 How to quote this

- **Correct:** “Off-plan stock sells at a higher rate per square metre because it is newer and better located.”

- **Correct:** “In the same building, a finished unit holds its value against an off-plan one.”
- **Wrong:** “Off-plan property is worth more than ready property.”
- **Wrong:** “Buyers pay a premium for off-plan.” They pay a premium for the buildings that happen to be sold off-plan.
- One caveat that no dataset can settle: the registry records the headline contract price. An off-plan price paid over three years of construction is not the same money as a resale price paid at once.

Appendix D. Price-band reconciliation

Bands were validated on all three views of the data. In each case the counts sum exactly to the row count, so every valid transaction belongs to exactly one band.

Each band includes its lower bound and excludes its upper one, so a sale of exactly AED 1,000,000 is counted in 1M – 2M, not in 500K – 1M. The top band is therefore AED 10M and above. The same rule is used by the dashboard.

Price band (AED)	Cleaned dataset	Share	Raw: residential units	Share	Raw: all sales	Share
< 500K	94,661	11.6%	123,584	13.3%	145,586	10.8%
500K – 1M	247,420	30.2%	286,294	30.8%	334,453	24.8%
1M – 2M	276,692	33.8%	303,580	32.7%	415,239	30.8%
2M – 3M	110,852	13.5%	117,162	12.6%	196,328	14.5%
3M – 5M	64,360	7.9%	68,128	7.3%	146,021	10.8%
5M – 10M	19,545	2.4%	21,783	2.3%	65,963	4.9%
> 10M	5,308	0.6%	7,958	0.9%	46,263	3.4%

Check	Result
Rows assigned to a band (cleaned)	818,838 of 818,838
Rows assigned to a band (raw residential units)	928,489 of 928,489
Rows assigned to a band (raw, all sales)	1,349,853 of 1,349,853
Null sale prices in raw sales	0
Zero or negative sale prices in raw sales	0
Duplicate transaction identifiers in raw sales	0
Sale prices below AED 1,000 in raw sales	39 (excluded from the cleaned dataset)
Cleaned dataset price range	AED 112,399 – 422,000,000
Cleaned sales of AED 10M or more	5,308 (matches the top band above)

Why the top band can read as zero on screen

The dashboard's sale-price slider starts at the 1st–99th percentile, which cuts off around AED 8M. Under that default the ">10M" band is empty — that is the filter, not the data. Unfiltered, the cleaned dataset holds 5,308 sales at AED 10M or more. The dashboard now displays a warning saying exactly this whenever a band is emptied by the filter.

Appendix E. Layout distribution reference

The nine layouts present in the dataset, with the quartiles behind the small-multiples chart in the Property section.

Layout	Transactions	Median size (m ²)	25th pct (AED/m ²)	Median (AED/m ²)	75th pct (AED/m ²)
Studio	189,729	39.9	10,341	14,845	19,086
1 B/R	343,595	73.4	10,222	14,854	20,647
2 B/R	211,748	119.8	10,955	15,758	22,669
3 B/R	64,318	179.5	11,171	17,010	24,433
4 B/R	7,614	315.1	12,037	19,943	32,451
5 B/R	950	581.1	9,087	18,325	36,581
6 B/R	82	896.6	19,069	32,254	40,551
7 B/R	4	1,535.2	30,273	51,432	67,074
PENTHOUSE	798	297.8	9,567	13,336	18,353

A caption that was corrected

The dashboard previously described this chart as showing that smaller units cost more per square metre. The data does not support that: the median rate rises with layout size. The caption was removed and replaced with what the data actually shows.

Appendix F. Every word, in plain English

Nothing on the Dubai dashboard requires a statistics background. Every term that is not everyday English is defined here, and the same definitions appear behind the ⓘ on each chart that uses them.

Amenity flag — A yes/no column recording whether a feature was noted for that unit. A “no” means it was **not recorded**, which is not the same as “definitely absent”.

Arima — A standard statistical forecasting model. It learns the pattern of a series from its own history and projects it forward. It uses no outside information.

Composition effect — When a difference between two groups turns out to be caused by the groups containing different kinds of thing, rather than by what you were measuring.

Confidence band — The range the model expects the true value to fall inside. A wide band means the model is not very sure.

Cumulative share — A running total, as a percentage. Where it crosses 80%, everything to the left of that point accounts for four-fifths of the market.

Distribution — The shape of the data — where most sales sit and how far the rest spread out.

Existing property — Already built and completed at the time it was sold. Also called resale or ready property.

Histogram — Sales sorted into equal-width buckets; the height of each bar is how many landed in that bucket.

Layout — The bedroom configuration: studio, 1 B/R, 2 B/R and so on.

Like-for-like — Comparing only properties that match on the things that matter — same area, same size of apartment, same year, same off-plan status — so the comparison is not secretly about something else.

Locality zone — A six-way grouping of Dubai areas carried in the dataset — Prime Central, Prime Waterfront, Urban Core, Suburban, Outer/Remote and Unknown.

Log scale — An axis where each step multiplies instead of adding (1M, 10M, 100M). Useful when the values run from small to enormous.

Mape — Mean absolute percentage error — on average, how far the model's past predictions were from what actually happened, in percent. Lower is better.

Median — Line every sale up from cheapest to dearest and take the one in the middle — half are below it, half above. Used instead of the average because a handful of enormous deals drags an average upwards and stops it describing anybody.

Off-plan — Bought from the developer before the building was finished.

Pareto chart — Bars sorted biggest first, with a line showing the running total, so you can see how few items make up most of the whole.

Partial year — A year the data does not cover in full. 2026 stops on 6 August, so eight months of it are being compared with somebody else's twelve.

Percentile — The 90th percentile is the value that 90% of sales come in below. The 1st and 99th are used here to trim off the few extreme deals at each end.

Price band — A price bracket, such as AED 1M – 2M. Every sale falls in exactly one band, so the bands add up to the whole market.

Quartile — Split the sales into four equal groups by price. The box in a box plot runs from the end of the first group to the end of the third — the middle half of the market.

Rate per m² — The price of one square metre: total price ÷ floor area. It lets a studio and a penthouse be compared fairly, because it takes size out of the picture.

Registration type — The Land Department's label for whether a sale was off-plan or existing. It is recorded on every transaction, so nothing here is guessed.

Rolling median — For each month, take that month together with the one before and the one after, and use the middle value of the three. It calms the zig-zag without moving the level or hiding anything — the real monthly figures are still there.

Total value — Every sale price in the group added together — how much money changed hands.

Treemap — Nested rectangles where the size of each box is proportional to the number it represents.

Violin — A smoothed outline of the distribution. Where it bulges, there are more sales at that price.

Whisker — The line reaching out of a box. It stretches to the furthest sale still inside the normal range; anything past it is an unusual deal.

Year-over-year — This year compared with the year immediately before it — never with a fixed starting year. So 2025 is measured against 2024, and 2024 against 2023.